A Second-Order Cone Programming (SOCP) Based Co-Optimization Approach for Integrated Transmission-Distribution Optimal Power Flow

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***Abstract*—The advancement of grid technology with high integration of distributed energy resources (DERs) has led to dynamic changes in the distribution network with a substantial impact on the transmission network. With this impact of DER integration, separate optimal power flow (OPF) analysis in transmission and distribution (TD) networks is not always adequate for the optimal operation of the power system. Hence, this paper proposes a coordinated TD-OPF co-optimization method with second-order cone programming (SOCP) to address this challenge. This co-optimization method provides a feasible solution for an integrated TD system, bridging the gap between transmission and distribution network challenges with renewable-based DERs through data exchange between the transmission and distribution networks. Moreover, each distribution network employs a distributed OPF (D-OPF) algorithm, which supports decentralized operation, thereby enhancing the reliability and scalability of the proposed OPF model for large TD networks. The goal is to reduce the overall system planning expenses, optimizing the microgrids in distribution networks and the major generation units within the transmission network. The efficacy of the proposed TD-OPF co-optimization approach has been verified through numerical analysis.**

***Index Terms*—Transmission-distribution Co-OPF, distributed OPF, angle & conic relaxation, SOCP-OPF.**

NOMENCLATURE

Abbreviations

ADMM	Alternating direction method of multipliers.
ADN	Active distribution networks.
APP	Auxiliary problem principle.
DER	Distributed energy resource.
DG	Distributed generator.
DSO	Distribution system operator.
OPF	Optimal power flow.
PCC	Point of common coupling.
SOCP	Second order cone programming.
TD	Transmission-distribution.
ToC	Time of convergence.
TSO	Transmission system operator.

Indices

ϵ_D	Tolerance limit of macro-iteration.
$\mathcal{N}_b$	Set of network's buses.
$\mathcal{N}_L$	Set of network's branches.
$\mathcal{N}_b^g$	Set of network's PV buses.
$\overline{(.)}$	Variables and parameters' maximum limits.
$\underline{(.)}$	Variables and parameters' minimum limits.
ϕ	Solution gap of the TD system.

Variables & Parameters

δ_{ij}	Voltage phase difference between buses $i\ \&;\ j \in \mathcal{N}_b$.
B_{ij}	The off-diagonal component's imaginary part of the network admittance matrix $(\mathbf{Y})$.
c_k^i	Generators' cost co-efficient at bus $i \in \mathcal{N}_b^g$, where $k = \{2, 1, 0\}$.
G_{ij}	The off-diagonal component's real part of the network admittance matrix $(\mathbf{Y})$.
I_{ij}	Current through branch $L_{ij} \in \mathcal{N}_L$.
l_{ij}	Square of the current magnitude through the branch $L_{ij} \in \mathcal{N}_L$.
P_i^d	Power demand (real) at bus $i \in \mathcal{N}_b$.
P_i^g	Power (real) at bus $i \in \mathcal{N}_b^g$.
P_{ij}	Power flow (real) through branch $L_{ij} \in \mathcal{N}_L$.
Q_i^g	Power (reactive) at bus $i \in \mathcal{N}_b^g$.
Q_i^d	Power demand (reactive) at bus $i \in \mathcal{N}_b$.
Q_{ij}	Power flow (reactive) through branch $L_{ij} \in \mathcal{N}_L$.
S_i^g	Power (apparent) at bus $i \in \mathcal{N}_b^g$.
S_i^d	Power demand (apparent) at the bus $i \in \mathcal{N}_b$.
S_{ij}	Power flow (apparent) through branch $L_{ij} \in \mathcal{N}_L$.
u_i	Voltage magnitude square at bus $i \in \mathcal{N}_b$.
V_i	Voltage at the bus $i \in \mathcal{N}_b$.
z_{ij}	Impedance of the branch $L_{ij} \in \mathcal{N}_L$.

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I. Introduction

Active distribution networks incorporate a large set of distributed generators (DGs) and renewable distributed energy resources (DERs) into the power distribution grid [1]. These active power sources make the distribution network complex, and the network needs more flexibility for stable network operation. The rising integration of DERs and DGs within their capacities necessitates coordinated operations between transmission and distribution networks. Traditionally, transmission and distribution networks have been managed and controlled independently. Although physically coupled, the transmission and distribution systems are managed by two distinct entities, the transmission system operator (TSO) and the distribution system operator (DSO), with minimal collaboration. As a result, there is minimal exchange of information regarding system conditions and control strategies between the TSO and DSO. Furthermore, the dynamic nature of the loads and the significant integration of DERs can lead to changes or even reversals in energy flow in specific feeders, posing challenges to traditional power system operations [2]. This integration of DERs could also complicate the utilization of flexible resources across the dual-layered grids of both networks. Due to these challenges, optimal coordination of transmission and distribution grids with DERs can provide a better and optimal point of operation for both networks [3]. However, there are significant challenges concerning the transmission-distribution network co-optimization for optimal power flow analysis.

- First, the existing AC-OPF model does not allow the decoupling of active and reactive power flows in active distribution grids. Disregarding the reactive power flow may result in errors in the OPF analysis.
- DERs integration is restricted by reverse power flow issues in the active distribution network. Large power injections into the distribution grids result in over-voltage [4]. So, OPF analysis is challenging within the network operation limits.
- Considering both the transmission and distribution grid, the scale of the problem is very extensive for the existing models for OPF analysis.

However, combining transmission and distribution grid models and centrally solving a single AC-OPF is impractical and challenging. Several decentralized algorithms for analyzing transmission-distribution networks have been reviewed in [5]. A heterogeneous decomposition method for economic dispatch in the coordinated transmission-distribution (TD) network is proposed in [6] for solving power flow in an integrated TD system. An iterative algorithm for microgrid planning is proposed in [7] as an alternative approach to co-optimizing TD networks to plan generation and transmission expansion in electric systems. In [8], a decomposition method is proposed to address the nonlinear aspects of the TD coordination analysis. Reactive power analysis of TD networks is done separately for both [7], [8], which might impact the solution accuracy.

The boundary buses are generally identified as points of common coupling (PCC) and are treated as PQ buses for the transmission grid, whereas the distribution grid considers the PCC as a PV bus. The high integration of DERs into distribution networks leads to over-voltage issues [4] and adds to the operational complexities of contemporary power distribution systems [9]. Furthermore, substantial integration of DERs into distribution networks also significantly affects transmission systems, as illustrated in [10]. So, the conventional power system models may fail to analyze the future grids [11].

The TD network analysis approach is addressed by various tools, including the Hierarchical Engine for Large-scale Infrastructure Co-Simulation (HELICS), the Framework for Network Co-Simulation (FNCS), the Integrated Grid Modeling System (IGMS), and GridSpice. However, these tools do not necessarily perform both transmission network (TN) and distribution network (DN) simulations or optimize them together [12]. A coordinated decentralized TD optimization method is discussed in [3], which used the branch flow equations for transmission and distribution networks and solved through quadratic approximation whose exactness might depend on the network architecture. A stochastic hierarchy for TD-OPF is developed and solved in parallel sub-problems with a nonlinear model of transformer in [13]. This method used second-order cone (SOC) relaxation in the transmission network and semidefinite programming (SDP) based relaxation in the distribution part. A co-simulation framework based on dynamic simulation in the transmission part and quasi-static time series analysis in the distribution part is developed in [14], considering real-time DERs' maximum output in a large network. The model in [15] used an SDP-based relaxation of power flow equations in the transmission network and a linear model for the distribution network. However, if the rank-1 constraint is not met, solutions obtained through SDP relaxation may become inexact, and linearization techniques suffer from inaccurate solutions [16]. Therefore, an exact and accurate integrated TD-OPF analysis is essential for modern power grid system OPF analysis.

OPF analysis in a power system is typically formulated using AC power flow with network operational constraints, and the OPF models are referred to as AC-OPF models. The AC-OPF problems are inherently NP-hard due to the nonlinear nature of the original AC power flow relationships [17]. Moreover, because of the non-convex nature of the original nonlinear OPF (NLP-OPF), globally optimal solutions cannot always be ensured [18]. To subside the nonlinearity, linear approximations are frequently employed; however, this compromises solution accuracy in OPF analysis [19]. Moreover, the linear OPF models do not consider the reactive power analysis. However, this article shows that reactive power is crucial at the PCC. However, convex OPF models (i.e., SOCP) [20]) are well known for their computational efficiency and ability to find globally optimal solutions and are widely adopted for analysis in power systems. Previous studies have demonstrated that conic and angle relaxation methods are exact in radial-type power distribution networks under specific conditions [21]. Moreover, if the network mesh cyclic conditions are satisfied, then the SOCP-OPF model provides an exact solution for meshed transmission networks as well [22]. So, this article considers the SOCP-based model for TD-OPF analysis.

Analysis of OPF in power distribution networks (DNs) primarily relies on centralized optimization techniques [23]. The proliferation of DERs leads to active distribution networks (ADNs) incorporating many active controllable equipment in the network [24]. This complexity makes the centralized OPF algorithms computationally challenging and sometimes infeasible. Furthermore, centrally controlled networks are vulnerable to cyber-physical attacks [25], and the entire network may fail due to a single-point failure. Considering these factors, distributed OPF analysis (D-OPF) is emerging as a viable option for decentralized operations in distribution networks. A well-known algorithm for distributed OPF is the Auxiliary Problem Principle (APP) technique [26]. The APP models break down the optimization problem into smaller sub-problems, each representing a section of the network system, and these sub-problems share variables at the PCC with adjacent areas of the network. Besides this, numerous D-OPF techniques utilizing the alternating direction method of multipliers (ADMM) algorithm have been explored in literature [27]. In ADMM-based D-OPF models, the convergence time (ToC) of the distributed OPF algorithm is determined by both micro and macro iterations. Despite employing multiple improvements, ADMM-based OPF models often require substantial iterations to converge; consequently, the ToC increases.

In [28], the computational efficiency and exactness of the SOCP-based D-OPF model are considered. It has been observed that with tight angle and conic relaxation, the model provides an exact solution for a radial distribution network with high integration of DERs. A notable research gap still exists where a method for a fast, exact, and globally optimal power flow model for TD-coupled networks even though SOCP-based OPF methods for separate analysis of transmission and distribution systems [22], [29] are exact. So, in this paper, a SOCP-based co-optimization model for coupled T & D networks is explored fully for integrated TD networks. Therefore, beyond our work in [28], this article proposes a TD-OPF model for transmission-distribution coupled networks utilizing the SOCP-based co-optimization model. The key contributions of the proposed TD-OPF model are as follows:

- The proposed TD-OPF model leverages the computational benefits of SOCP in meshed transmission networks and provides an exact solution considering an approximated voltage phase angle to satisfy the angle cyclic constraints in the mesh cycles.
- The proposed TD-OPF model can provide an exact solution with high integration of DERs in the distribution network. The model can control the reactive power constraints at the PCC and the network, thus providing a feasible solution within the operational limit of the network voltage.
- The TD-OPF model leverages the computational benefits of the SOCP D-OPF model for distribution networks. The D-OPF model considers both the voltage and angle, while the model in [28] considers only the bus voltage. The operational constraints on both the bus voltage amplitude and phase angle provide greater precision and help to satisfy the cyclic constraints in the transmission network. With this D-OPF model on the distribution side, the proposed TD-OPF model is applicable to extensive integrated transmission-distribution networks.

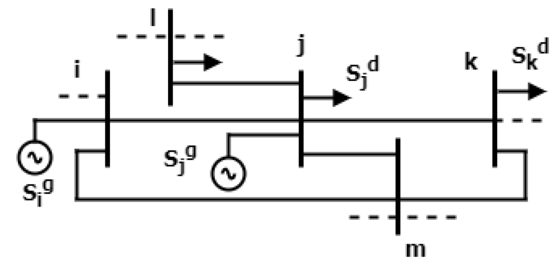

Fig. 1. One-line diagram of a simple power grid.

The rest of the paper is organized as follows. The SOCP-based OPF methodology and the co-optimization OPF methodology are discussed in Section II. Section III discusses the formulation and exactness of transmission-distribution co-optimization OPF analysis. Finally, Section IV concludes the article by highlighting the directions for future research.

II. Proposed Methodology

A power grid system is a combination of transmission and distribution systems. The distribution system is normally radial, while the transmission grid is mainly meshed in structure. To analyze the power grid system, $\mathcal{N}_b$ is the set of network buses, and the set of network branches is defined as $\mathcal{N}_L$. The power flow from bus $i \in \mathcal{N}_b$ to bus $j \in \mathcal{N}_b$ is:

$$S_{ij} = V_i I_{ij}^* \tag{1}$$

where S_{ij} is the apparent power flow and I_{ij} current flow through $L_{ij} \in \mathcal{N}_L$. Further, with the equivalent half lump shunt defined as $y_j = g_j + jb_j$, the apparent power balance at bus $j \in \mathcal{N}_b$ is represented as follows:

$$S_j^g - S_j^d = \sum_{j\to k} S_{jk} - \sum_{i\to j}(S_{ij} - \mathbf{z}_{ij}|I_{ij}|^2) + y_j^*|V_j|^2 \tag{2}$$

Real and reactive power flow at $L_{ij} \in \mathcal{N}_L$ are represented as:

$$P_{ij} = -G_{ij}V_i^2 + G_{ij}V_iV_j\cos(\delta_{ij}) + B_{ij}V_iV_j\sin(\delta_{ij}) \tag{3}$$

$$Q_{ij} = B_{ij}V_i^2 - B_{ij}V_iV_j\cos(\delta_{ij}) + G_{ij}V_iV_j\sin(\delta_{ij}) \tag{4}$$

where $\mathbf{Y}_{ij} = G_{ij} + jB_{ij}$ is the off-diagonal element of the impedance matrix $\mathbf{Y}$. δ_i is phase of voltage at bus $j \in \mathcal{N}_b$ and $\delta_{ij} = \delta_i - \delta_j$. Further, from (3) and (4):

$$V_iV_j\sin\delta_{ij} = \frac{B_{ij}P_{ij} + G_{ij}Q_{ij}}{G_{ij}^2 + B_{ij}^2} \tag{5}$$

As the bus $i \in \mathcal{N}_b$ and bus $j \in \mathcal{N}_b$ are two adjacent buses connected in the network, then $V_iV_j \approx 1$. Further, if two adjacent bus voltage phase difference is $\sin(\delta_{ij}) \approx \delta_{ij}$, then:

$$\delta_{ij} = \frac{B_{ij}P_{ij} + G_{ij}Q_{ij}}{G_{ij}^2 + B_{ij}^2} \tag{6}$$

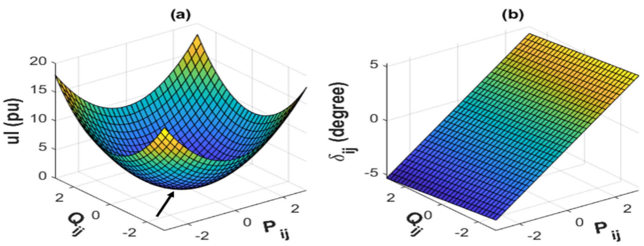

Fig. 2. (a) Conic-convex solution space; (b) Feasible δ_{ij}.

For any mesh in power system networks, the cyclic constraints are defined as follows:

$$\sum_{(i,..x)\in\mathcal{C}} \delta_{ij} + \delta_{jk} + ... + \delta_{xi} = 0 \tag{7}$$

where bus i and $x \in \mathcal{N}_b$ within a particular mesh cycle ($\mathcal{C}$).

A. Power Flow Relaxation

1) Angle Relaxation: For the SOCP model, the power flow relations relax the phase angles and consider the magnitude only. After relaxation, the power flow relations $S_{ij} = V_i I_{ij}^*$ and the voltage drop $V_j = V_i - \frac{z_{ij}S_{ij}^*}{V_i^*}$ across $L_{ij} \in \mathcal{N}_L$ are converted as follows:

$$S_{ij}^2 = u_i l_{ij} \tag{8}$$

$$u_j = u_i - 2\left(r_{ij}P_{ij} + x_{ij}Q_{ij}\right) + \left(r_{ij}^2 + x_{ij}^2\right) l_{ij} \tag{9}$$

where, $|I_{ij}|^2 = l_{ij}$ & $|V_i|^2 = u_i$. The apparent power balance at the bus $j \in \mathcal{N}_b$ is:

$$S_j^g - S_j^d = \sum_{j\to k} S_{jk} - \sum_{i\to j}(S_{ij} - z_{ij}l_{ij}) + y_j u_j \tag{10}$$

where S_j^g represents the apparent power injected and S_j^d denotes the apparent load connected at bus $j \in \mathcal{N}_b$. Decomposing (10) into its components of real and reactive power:

$$P_j^g - P_j^d = \sum_{j\to k} P_{jk} - \sum_{i\to j}(P_{ij} - r_{ij}l_{ij}) + g_j u_j \tag{11}$$

$$Q_j^g - Q_j^d = \sum_{j\to k} Q_{jk} - \sum_{i\to j}(Q_{ij} - x_{ij}l_{ij}) + b_j u_j \tag{12}$$

where $z_{ij} = r_{ij} + jx_{ij}$; r_{ij} and x_{ij} are the resistance and reactance of the line $L_{ij} \in \mathcal{N}_L$ respectively. $S_{ij} = P_{ij} + jQ_{ij}$, $y_j = g_j + jb_j$, $S_j^g = P_j^g + jQ_j^g$, and $S_j^d = P_j^d + jQ_j^d$.

2) Conic Relaxation: The power flow relation in (8) is still non-convex because of its second-order quadratic form. The proposed SOCP OPF model creates a convex solution space through conic relaxation (13), and this conic-convex space is illustrated in Fig. 2(a).

$$u_i + l_{ij} \geq \left\| \begin{matrix} 2P_{ij} \\ 2Q_{ij} \\ u_i - l_{ij} \end{matrix} \right\|_2 \tag{13}$$

If a function is a conic convex, then any local minimum of that function is the global minimum. So, the global minimum is denoted with an arrow ($\nearrow$) sign in 2(a). So, if the OPF model can be converted to conic convex, it will always provide the optimal global solution.

3) SOCP D-OPF & Co-Optimization-Based TD-OPF Model: For the tightness of the relaxations, only convex objective functions $f(x)$ (i.e., loss minimization as, $\min \sum r_{ij}l_{ij}$ and generation cost minimization as, $\min \sum [c_2^i(p_i^g)^2 + c_1^i p_i^g + c_0^i])$ are considered in this article with $c^i \geq 0$. Here, $c_2^i(\$/MWh^2)$, $c_1^i(\$/MWh)$ and $c_0^i(\$/h)$ are the generator's cost coefficients. For OPF analysis, the network operational constraints are defined as follows:

$$\text{s.t.} \begin{cases} \underline{u}_i \leq u_i \leq \overline{u}_i \\ \underline{\delta}_i \leq \delta_i \leq \overline{\delta}_i \\ \underline{P}_i^g \leq P_i^g \leq \overline{P}_i^g \\ \underline{Q}_i^g \leq Q_i^g \leq \overline{Q}_i^g \\ l_{ij} \leq \overline{l}_{ij} \end{cases} \tag{14}$$

where $\underline{(.)}$ and $\overline{(.)}$ are the minimum and maximum limits of parameters and variables. For an objective function $f(x)$, the SOCP D-OPF model for the distribution network is: $\min \ f(x)$ subject to (9) and (11)–(14). As the objective functions and the constraints are both convex, the optimization problem will provide a globally optimal solution. For meshed networks, the loops of a network are determined using the algorithm presented in [22]. Along with the OPF model for the distribution network, cyclic constraints (7) in the meshed transmission network are also considered for the proposed TD-OPF model.

B. SOCP D-OPF Model for Distribution Network

In the SOCP-based D-OPF model proposed in [28], the model denotes the area towards the sub-station as the upstream area (UA), while the radial end areas are denoted as downstream areas (DA). A data set $\mathbf{z}_1$ comprising P_i^g and Q_i^g from the DA is transmitted to the UA. Simultaneously, $\mathbf{z}_2$, containing V_i and δ_i from the UA, is transmitted to the DA for utilization in the subsequent macro iteration at the point of common coupling (PCC). The simulation converges if the $\mathbf{z}_2$ from subsequent macro iterations is lower than a certain margin (ϵ_D) as $|\mathbf{z}_{2,PCC}^{t+1} - \mathbf{z}_{2,PCC}^t| \leq \epsilon_D$. The desired values of $\mathbf{z}_1$ and $\mathbf{z}_2$ are determined utilizing the SOCP-based OPF model proposed in (9) and (11)–(14). The OPF solution area is enclosed with a cone defined by the relaxed second-order conic relation derived in (13). Being a convex model, the proposed model guarantees a global optimal value of $\mathbf{z}_1$ and $\mathbf{z}_2$. For analysis, we have utilized the MOSEK solver for the SOCP-OPF simulation. The exactness of the SOCP-based OPF model is illustrated later in this article.

If a distribution network is divided into $N_{\mathcal{A}}$ numbers of areas, the proposed SOCP D-OPF model's macro iteration conditions are illustrated in the Algorithm 1. Furthermore, the performance of this SOCP D-OPF model is compared with a traditional ADMM-based D-OPF model as in [30], [31]. For the ADMM model, the overall convergence time (ToC) is influenced by both micro and macro iterations. Micro iterations refer to the steps required for achieving convergence within each local area. Meanwhile, macro iterations involve the exchange of variables

Algorithm 1: Macro Iterations of the D-OPF.

$\epsilon_D = 10^{-6}$, $\Delta\mathbf{z} = 1$ and $m = 1$
while $\Delta\mathbf{z} \geq \epsilon_D$ **do**
  if *Area* $\mathcal{A}_i$ *is either UA or DA* **then**
    if $m = 1$ **then**
      $\mathbf{z}_1$ & $\mathbf{z}_2 \leftarrow$ initialization
    end
    if $m > 1$ **then**
      $\mathbf{z}^{t+1}_{1,UA(\mathcal{A}_i)} \leftarrow \mathbf{z}^{t}_{1,DA(\mathcal{A}_{i+1})}$ *(For UA)
      $\mathbf{z}^{t+1}_{2,DA(\mathcal{A}_{i+1})} \leftarrow \mathbf{z}^{t}_{2,UA(\mathcal{A}_{i})}$ *(For DA)
    end
    $\Delta\mathbf{z}_i = |\mathbf{z}^{t+1}_2 - \mathbf{z}^{t}_2|$
  end
  if *Area* $\mathcal{A}_i$ *is both UA & DA* **then**
    if $m = 1$ **then**
      $\mathbf{z}_1$ & $\mathbf{z}_2 \leftarrow$ initialization
    end
    if $m > 1$ **then**
      $\mathbf{z}^{t+1}_{1,UA(\mathcal{A}_i)} \leftarrow \mathbf{z}^{t}_{1,DA(\mathcal{A}_{i+1})}$
      $\mathbf{z}^{t+1}_{2,DA(\mathcal{A}_{i})} \leftarrow \mathbf{z}^{t}_{2,UA(\mathcal{A}_{i-1})}$
    end
    $\Delta\mathbf{z}_i = |\mathbf{z}^{t+1}_2 - \mathbf{z}^{t}_2|$
    *Two $\Delta\mathbf{z}_{\mathcal{A}_i}$ from two PCCs
  end
  $\Delta\mathbf{z} = \max\{\Delta\mathbf{z}_1, \Delta\mathbf{z}_2.....\Delta\mathbf{z}_{N_{\mathcal{A}}-1}\}$
  $m \leftarrow m + 1$
end

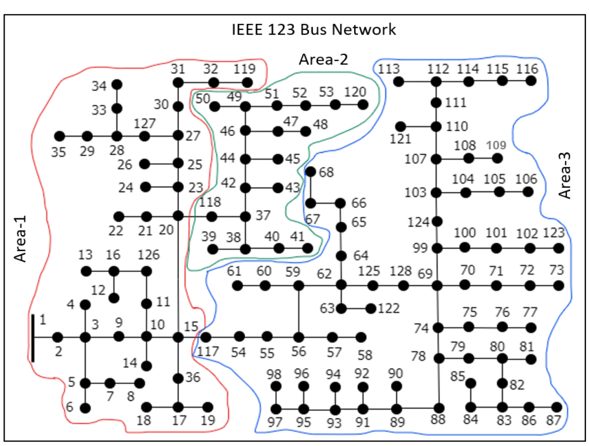

Fig. 3. IEEE 123-bus network with bus reconfiguration.

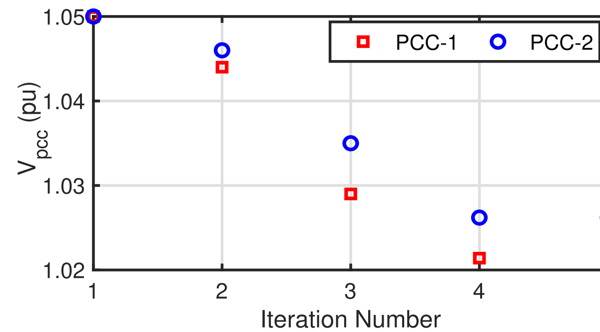

Fig. 4. PCC voltage for macro-iterations in D-OPF analysis.

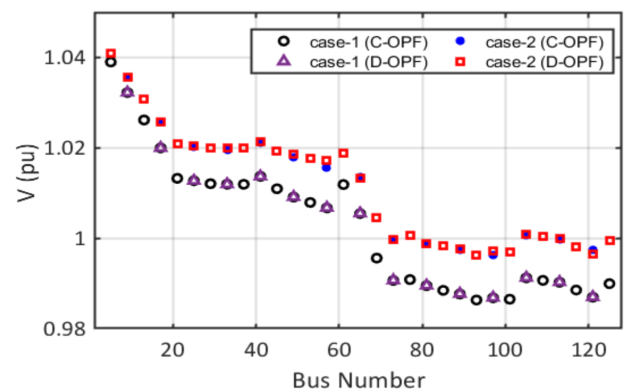

Fig. 5. Voltage profile from SOCP D-OPF & C-OPF.

TABLE I
DERS' POWER INJECTION INTO THE 123-BUS NETWORK

Bus No.	C-OPF		D-OPF	
	$P_g(kw)$	$Q_g(kvar)$	$P_g(kw)$	$Q_g(kvar)$
11	13.30	8.82	13.30	8.85
30	13.30	8.80	13.35	8.80
60	6.75	4.45	6.75	4.44
67	46.75	30.9	46.75	30.85
78	13.30	8.80	13.30	8.83
89	13.30	8.80	13.30	8.83
50	70.00	46.42	70.00	46.45
120	13.30	8.80	13.30	8.80

and operational data between the decision-making agents. The following subsection will present a simulation analysis to assess the efficiency of the proposed D-OPF model.

1) Analysis on Distribution Networks: The analysis of the SOCP D-OPF model is conducted using IEEE 123-bus and 8500-bus network systems for experimental validation. In the experimental setup, a large number of DERs (30% of the connected total load to the network) is integrated into the standard IEEE 123-bus network. The network is divided into three areas at its switching buses, as illustrated in Fig. 3. Within this configuration, PCC-1 is the connection point for Area-1 and Area-2, whereas PCC-2 serves as the connection point between Area-1 and Area-3. The voltage profile at the PCCs in the 123-bus network with DERs is illustrated in Fig. 4 for macro-iterations of the D-OPF model. This illustration confirms that the D-OPF model achieves convergence by the fifth iteration when DERs are present in the system. The difference in voltages obtained from the D-OPF model and the centralized SOCP-OPF (C-OPF) model is minimal, under 0.01%, for both case-1 & case-2, and the voltage profile of the network is presented in Fig. 5. Here, case-1 represents the network's base case scenario, while case-2 incorporates the effects of DER integration. The injection of power from the DERs is shown in Table I, and the network's percentage loss is illustrated in Fig. 6. Subsequently, the D-OPF model's applicability is extended to assess its performance in a significantly extensive network (i.e., the IEEE 8500-bus network).

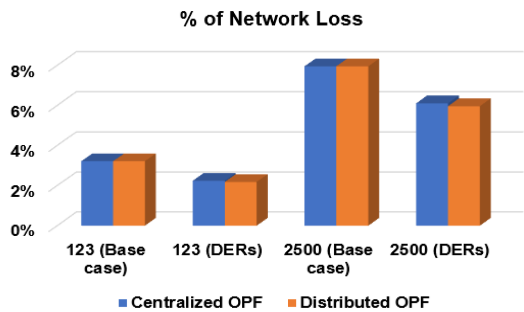

Fig. 6. % of network loss in C-OPF and D-OPF analysis.

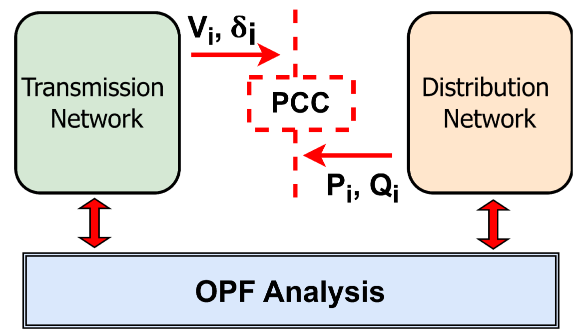

Fig. 7. TD co-optimization-based OPF configuration.

TABLE II
TIME OF CONVERGENCE (TOC) FOR C-OPF & D-OPF

Test case	C-OPF ToC (sec)	D-OPF		
		Area division	ToC (sec)	
	SOCP		SOCP	ADMM
123-bus (case-1)	0.25	3	0.88	136
123-bus (case-2)	0.30	3	0.52	93
2522-bus (case-1)	1.06	3	2.68	681
2522-bus (case-2)	1.53	3	3.12	476
2522-bus (case-1)	1.06	6	7.56	N/A
2522-bus (case-2)	1.53	6	8.66	N/A

To analyze D-OPF in the IEEE-8500 bus network, an equivalent single-phase 2522-node system is developed from the IEEE 8500-bus distribution network. The 2522-bus network is divided into several areas for conducting the experimental study for the two cases where the network is divided into three and six areas. The comparison between the D-OPF and the C-OPF analysis shows a voltage difference of less than 0.01% for both three & six area divisions, respectively. The percentage of network losses for both the 2522-node network and the IEEE 123-bus network is depicted in Fig. 6, indicating that network losses are higher in the more extensive 2522-bus network compared to the 123-bus network. Table II compares the time required to reach convergence (ToC) between the SOCP D-OPF and the ADMM-OPF models. In this table, case-1 represents the network's base case, while case-2 includes the integration of DERs into the network. Despite the SOCP D-OPF model having a higher ToC than the SOCP C-OPF due to its macro-iterations, it is significantly more efficient than the ADMM-based D-OPF model. Moreover, the D-OPF model has substantial scalability advantages over the C-OPF, being adaptable to any division number in an extensive power system network (i.e., the IEEE 8500-bus network) where the C-OPF may not be successful for a feasible solution. The subsequent sections of this paper will provide a detailed illustration of the proposed co-optimization-based integrated transmission-distribution OPF (TD-OPF) model.

III. Co-Optimization of Integrated Transmission and Distribution Networks

The transmission and distribution networks are connected to the PCC substation. For the TD-OPF co-optimization, a data set of $\mathbf{z}_1^{TD}$ (i.e., P_i, Q_i) from the distribution network is passed to the transmission network, and $\mathbf{z}_2^{TD}$ (i.e., V_i, δ_i) from the transmission network to the distribution network for each macro-iteration at the PCC. Here, P_i, Q_i represents the equivalent PCC power for a distribution network transmitting from the transmission network, which will be considered a fixed load at the transmission terminal for the transmission network simulation. Moreover, V_i, δ_i are the voltage magnitude and voltage angle of the PCC for the transmission network simulation, which will act as a reference for the distribution network simulation. The data transition between the transmission and distribution network is illustrated in Fig. 7. The arrow signs represent that after transmission network simulation, the V_i, δ_i will be passed to the PCC for distribution network simulation, and P_i, Q_i will be passed for the subsequent macro-iteration of the transmission network. The proposed SOCP TD-OPF converges when $|\mathbf{z}_2^{TD,t} - \mathbf{z}_2^{TD,t-1}| \leq \epsilon$. The algorithm is illustrated in detail in Fig. 8. At the beginning of the co-optimization, the model acquires the network data (i.e., line impedance, generation capacity) followed by the DERs' and network load status. The distribution side network is analyzed with the D-OPF algorithm in 1 with the reference of the voltage and phase angle from the connected PCC for that distribution feeder. Then, the total real and reactive power followed to the network from the sub-station is considered the static load for the respective connected bus of the transmission network at that PCC. If the voltage difference of the macro-iteration is under a certain value, then the D-OPF converges.

Similarly, the model looks for any network loading condition of DERs' status change to analyze the network for the next point of optimal operation. The proposed model incorporates an approximated angle recovery relation in (6) for the cyclic constraints in the meshed transmission network. One of the significant challenges for the co-optimization of the SOCP TD-OPF model is the exactness of the solution. The exactness of the SOCP-OPF model in a meshed transmission network is illustrated in [22] with an angle recovery from the relaxation. The next section of this article demonstrates the exactness of the proposed TD-OPF model with the proposed approximated angle recovery model.

A. Exactness of the Proposed TD-OPF Model

In this section of the paper, we have focused on the exactness of the transmission network. For an exact solution from the proposed TD-OPF model, the angle and conic relaxations should be tight for both the transmission and distribution networks. It has already been established that for radial-type networks,

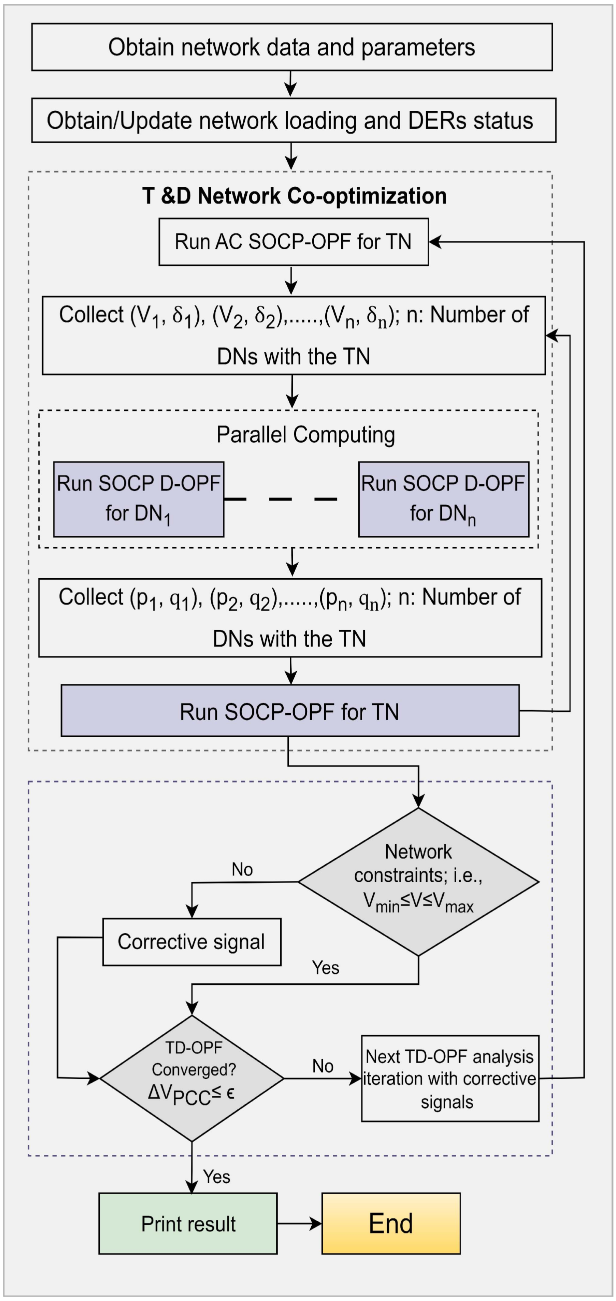

Fig. 8. Proposed SOCP TD-OPF co-optimization algorithm.

under certain conditions, both conic and angle relaxations are exact [29], [32]. The conic relaxation is tight for meshed transmission networks, but the angle relaxation is not always. To evaluate the exactness of the solution, a quantitative metric known as the solution gap, denoted by $\sigma = |S_{ij}^2 - u_i l_{ij}|$ is introduced, which will be used throughout this article.

In the SOCP TD-OPF model, the objective function $f(x)$ is considered convex and strictly increasing concerning the current flow I_{ij}. We denote the set of optimal feasible solutions obtained from the proposed model as $\tilde{\Psi} = (\tilde{u}, \tilde{P}g, \tilde{S}ij, \tilde{l}_{ij})$. If there is another solution $\hat{\Psi} = (\hat{u}, \hat{P}_g, \hat{S}_{ij}, \hat{l}_{ij})$ exists in the solution space, which is exact and satisfies the network cyclic constraints, where $\hat{l}_{ij} = \tilde{l}_{ij} - \xi, \hat{S}_{ij} = \tilde{S}_{ij} - \xi\mathbf{z}_{ij}, \hat{u}_i = \tilde{u}_i, \hat{S}_i^g = \tilde{S}_i^g, \hat{S}_i^d = \tilde{S}_i^d + \xi\mathbf{z}_{ij}$ and $\hat{S}_j^d = \tilde{S}_j^d$ for a $\xi \geq 0$. If the objective value $f(\hat{\Psi})$ be less than $f(\tilde{\Psi})$, due to $\hat{l}ij = \tilde{l}ij - \xi$, this situation challenges the optimality of $\tilde{\Psi}$ as derived from the proposed TD-OPF model. The tightness and global optimality of the proposed TD-OPF co-simulation model is achieved when no other solution set is within the solution space that is lower than $\tilde{\Psi}$, and all cyclic constraints are adequately met. The exactness of the proposed model is further illustrated and analyzed through the following remark in this section.

Remark 1: As, $\hat{\Psi}$ is an exact solution; the solution gap $\hat{\phi} = \hat{u}_i\hat{l_{ij}} - \hat{S_{ij}^2} \approx 0$. So, the solution from the TD-OPF model ($\tilde{\Psi}$) is exact if $\tilde{\Psi} = \hat{\Psi}$.

Proof: As $\tilde{\Psi}$ is a feasible solution within the conic space $\tilde{u}_i\tilde{l}_{ij} - \tilde{S}_{ij}^2 \geq 0$. Then,

$$\begin{aligned} &\tilde{u}_i\tilde{l}_{ij} - \tilde{S}_{ij}^2 \\ &\quad \Rightarrow \hat{u}_i(\hat{l}_{ij} + \xi) - (\hat{S}_{ij} + z_{ij}\xi)^2 \\ &\quad \Rightarrow \hat{u}_i\hat{l}_{ij} - \hat{S}_{ij}^2 + \xi\left[\hat{u}_i - z_{ij}^2\xi - 2\hat{S}_{ij}z_{ij}\right] \end{aligned}$$

If $\xi = 0$; then $\tilde{u}_i\tilde{l}_{ij} - \tilde{S}_{ij}^2 = \hat{u}_i\hat{l_{ij}} - \hat{S_{ij}^2}$. Thus the Solution gap $\tilde{\phi} \approx 0$ for $\tilde{\Psi}$ and the solution from the proposed model will be exact for the meshed network. The solution's exactness is achieved when cyclic angle constraints are satisfied in the meshed transmission network section.

The globally optimal and exact solution set $\hat{\Psi}$ meets the network's cyclic angle constraints. Therefore, for a meshed cyclic network that includes a branch $L_{ij} \in \mathcal{N}_L$, the cyclic angle constraint satisfaction is achieved as follows:

$$\begin{aligned} &\frac{B_{ij}\hat{P_{ij}} + G_{ij}\hat{Q_{ij}}}{(G_{ij}^2 + B_{ij}^2)} + \frac{B_{jk}P_{jk} + G_{jk}Q_{jk}}{(G_{jk}^2 + B_{jk}^2)} \\ &\quad + \cdots + \frac{B_{j'i}P_{j'i} + G_{j'i}Q_{j'i}}{(G_{j'i}^2 + B_{j'i}^2)} = 0 \end{aligned} \tag{15}$$

In the proposed SOCP TD-OPF model, the cyclic angle constraints are applied for the solution $\tilde{\Psi}$. In a mesh network that includes branch $L_{ij} \in \mathcal{N}_L$, the solution set $\tilde{\Psi}$ satisfies the constraint as follows:

$$\begin{aligned} &\frac{B_{ij}\tilde{P_{ij}} + G_{ij}\tilde{Q_{ij}}}{(G_{ij}^2 + B_{ij}^2)} + \frac{B_{jk}P_{jk} + G_{jk}Q_{jk}}{(G_{jk}^2 + B_{jk}^2)} \\ &\quad + \cdots + \frac{B_{j'i}P_{j'i} + G_{j'i}Q_{j'i}}{(G_{j'i}^2 + B_{j'i}^2)} = 0 \\ &\Rightarrow \frac{B_{ij}\hat{P_{ij}} + G_{ij}\hat{Q_{ij}} + \xi(B_{ij}r_{ij} + G_{ij}x_{ij})}{(G_{ij}^2 + B_{ij}^2)} \\ &\quad + \frac{B_{jk}P_{jk} + G_{jk}Q_{jk}}{(G_{jk}^2 + B_{jk}^2)} + \cdots\ldots \\ &\quad + \frac{B_{j'i}P_{j'i} + G_{j'i}Q_{j'i}}{(G_{j'i}^2 + B_{j'i}^2)} = 0 \end{aligned} \tag{16}$$

The proposed TD-OPF model is a convex model and satisfies the angle cyclic constraints. Therefore, the optimal solution

TABLE III
SUMMARY OF OBSERVATIONS

Methods	Feasibility	Optimality	Accuracy	Scalability	ToC	Comments
D-OPF						
APP	Feasible	Global Optimal	Exact	Scalable	Very Slow	Feasible and exact but very slow for communications delay and feasibility depends on solution methodology
ADMM	Feasible	Global Optimal	Exact	Scalable	Slow	Feasible and exact but convergence time depends on Lagrange multipliers and feasibility depends on solution methodology
SOCP D-OPF	Feasible	Global Optimal	Exact	Scalable	Fast	Exact, fast, and scalable
TD-OPF						
NLP TD-OPF	Depends on systems	Global Optimal	Exact	Not scalable	Very Slow	Exact but feasibility and scalability depends on systems and very slow
DC TD-OPF	Feasible	Optimal	Not accurate	Scalable	Fast	Feasible, fast and optimal but not accurate
SOCP TD-OPF	Feasible	Global Optimal	Exact	Scalable	Fast	Exact, fast, and scalable

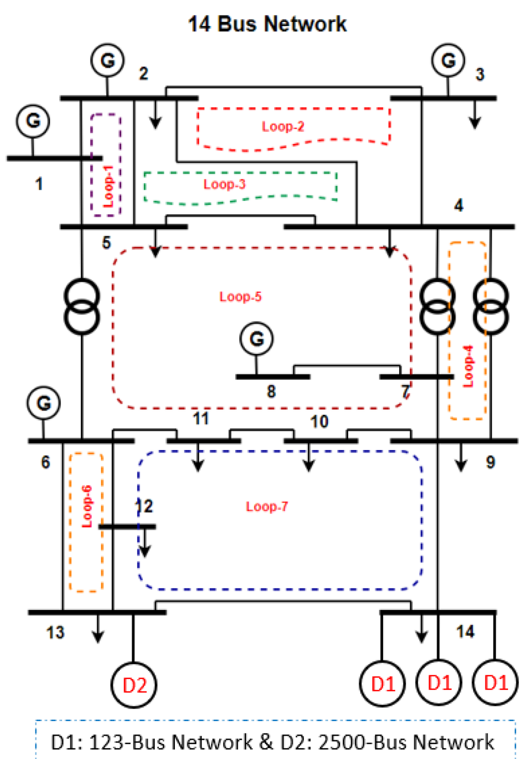

Fig. 9. Transmission-Distribution (TD) network configuration.

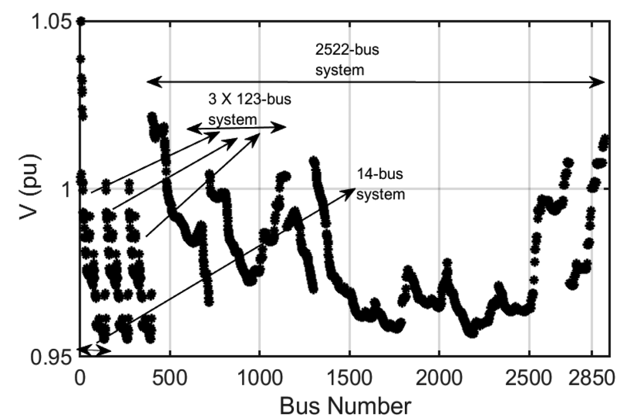

Fig. 10. Voltage profile for the TD network system; first IEEE 14-bus network, then 3× IEEE 123-bus network, and then, IEEE 2522-bus network.

$\tilde{\Psi}$ is the globally optimal solution, illustrated in this article's Appendix A. Further, in both (15) and (16), the cyclic constraints of the network analysis are satisfied, and $f(\tilde{\Psi})$ is the globally optimal solution, so, $\xi = 0$ and the solution from the proposed TD-OPF model is exact. ■

B. Analysis of TD-OPF Co-Optimization

In this study, we conduct a SOCP TD-OPF analysis that includes both the transmission and distribution networks, adopting a convex objective function similar to the approach in the D-OPF model. For computational processing, we employ the MOSEK solver integrated with MATLAB. The convexity of the proposed SOCP TD-OPF model enables it to reliably generate feasible solutions for the interconnected TD networks, even with significant integration of DERs on the distribution end. In this framework, the transmission network functions as the upstream or master areas, whereas the connected distribution networks are treated as the downstream or slave areas. The D-OPF model is also applied within the distribution networks as part of the TD-OPF model. Illustrating with a test case, as depicted in Fig. 9, the scenario includes an IEEE 14-bus transmission network connected to three IEEE 123-bus networks and an extensive IEEE 2500-bus network. The 123-bus networks are connected to bus number 14, and the 2500-bus network is connected to bus number 13 of the IEEE 14-bus network, which shows the model's capability to handle complex TD network configurations.

For the distribution networks, high penetration of DERs is considered with both 123-bus and 2500-bus networks. The voltage profile is illustrated in Fig. 10. It shows that for an optimal feasible solution, the voltage level is within the operational limit of $[0.95, 1.05]$ p.u. The primal error (α_p) and the dual error (α_d) convergence are illustrated in Fig. 11, where, (a), (b), and (c) are for the 14-bus, 123-bus and 2522-bus networks respectively. The OPF model converges in less than 10 iterations for all the networks. This proves the capability of very fast convergence to the solution of the proposed SOCP TD-OPF model.

C. Exactness and Comparative Study of TD-OPF Models

The impact of the cyclic constraints on the exactness of the SOCP TD-OPF model is greater in large transmission networks with a larger number of mesh cycles. The model is analyzed

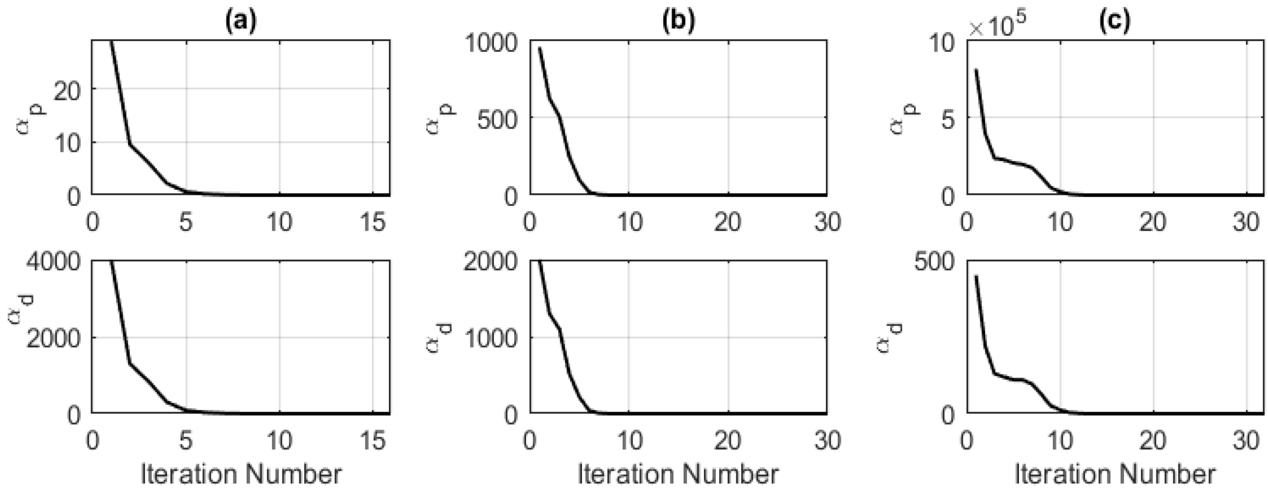

Fig. 11. Convergence of the SOCP TD-OPF model for the T&D system in Fig. 9.

using different transmission and distribution networks (i.e., TD in combination with IEEE 9, 14, 118-bus transmission & IEEE 33, 123, 2522-bus distribution networks). The result shows that if cyclic constraints are satisfied, then the solution gap (σ) is lower than if not satisfied for the OPF model. If the solution is not exact at the transmission side, then this solution provides a wrong reference point for the distribution side through the PCC. So, the solution from the TD-OPF model is not the point of optimal operation of the network. The integrated TD-OPF is compared with a nonlinear OPF model (NLP TD-OPF) with the Sequential Quadratic Programming (SQP) method as proposed in [16]. For the NLP-OPF model, the transmission network is connected with the distribution network through a step-down transformer at the PCC to match the voltage level. Besides the NLP TD-OPF model, the proposed SOCP TD-OPF model is also compared with DC TD-OPF as well. A comparative study on the TD-OPF model is summarized in Table III, and the study shows that the proposed SOCP TD-OPF model is scalable for large networks for exact solutions with fast convergence. Besides the TD-OPF models, the table also illustrates a comparative study of different models for distributed D-OPF analysis in distribution networks.

IV. Conclusion

This article introduces a transmission-distribution co-optimization planning model (SOCP TD-OPF) that accounts for network operational reliability and economic power dispatch in modern power grid systems. The proposed TD-OPF model harnesses the computational advantages of the SOCP algorithm and is thus applicable for extensive power networks with high integration of DERs in the grid. Based on the analysis, the proposed TD-OPF model can provide an exact solution. It effectively manages voltage levels and real and reactive power flow throughout the network, ensuring a solution that aligns with network operational limits. This article also conducts a comprehensive study of the D-OPF model for the distribution grid and the proposed TD-OPF model of TD networks with the conventional OPF models. The analysis reveals that the relaxations and approximations used to achieve OPF convergence are tight, providing a globally optimal solution for convex objective functions. Considering the computational efficiency and accuracy of the SOCP approach, the SOCP TD-OPF model will be considered for real-time OPF analysis and DER control for future research.

Appendix A

This section illustrates the global optimality of the proposed convex OPF model. Lets assume $f(\tilde{\Psi}) \geq f(\hat{\Psi})$ for for the two solution sets $\tilde{\Psi}$ and $\hat{\Psi}$. For any convex objective functions, it can be shown that:

$$\gamma f(\hat{\Psi}) + (1-\gamma) f(\tilde{\Psi}) \geq f\left(\gamma\hat{\Psi} + (1-\gamma)\tilde{\Psi}\right) \tag{17}$$

where $\gamma \in [0, 1]$. Then;

$$\begin{aligned} \gamma f(\tilde{\Psi}) + (1-\gamma) f(\tilde{\Psi}) &\geq \gamma f(\hat{\Psi}) + (1-\gamma) f(\tilde{\Psi}) \\ \Rightarrow f(\tilde{\Psi}) &\geq \gamma f(\hat{\Psi}) + (1-\gamma) f(\tilde{\Psi}) \end{aligned} \tag{18}$$

Considering (17) and (18):

$$f(\tilde{\Psi}) \geq f\left(\gamma\hat{\Psi} + (1-\gamma)\tilde{\Psi}\right) \tag{19}$$

If $f(\tilde{\Psi})$ is the optimal solution then for any other solution set $f(\Psi) > f(\tilde{\Psi})$. However, this conflicts with (19). Only $\tilde{\Psi} = \hat{\Psi}$; satisfies the both conditions. Therefore, a convex OPF model with a convex objective function always provides a globally optimal solution.

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